TOP-Q — Day 2 worked notes

Chiral symmetry, winding number, edge states and localisation length

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Same rules as Day 1: re-derive Sections 1–6 on paper without looking. Every result is labelled [Derived], [Verified-by-test], [Simulated], [Literature] or [Hypothesis]. Conventions are unchanged: $d\boldsymbol{a}/dz = iH\boldsymbol{a}$, b = propagation-constant shift, chain starts on A, $r \equiv \kappa_1/\kappa_2$.

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1 Chiral (sublattice) symmetry

In simple words: in an SSH array, an A waveguide only couples to B waveguides, and a B waveguide only couples to A waveguides. Nothing couples A to A or B to B, and no waveguide has its own extra detuning. That "only A–B" rule is the whole symmetry.

1.1 The operator

Define $\Gamma = \mathrm{diag}(+1, -1, +1, -1, \dots)$: $+1$ on A guides, -1 on B guides. $\Gamma^2 = \mathbb{1}$.

1.2 Proof that $\Gamma H \Gamma = -H$ [Derived]

$(\Gamma H \Gamma)_{mn} = \gamma_m H_{mn} \gamma_n$, with $\gamma = \pm 1$.

- If H_{mn} joins A to B: $\gamma_m \gamma_n = -1$, so the entry flips sign.
- If H_{mn} were a diagonal (on-site) term or joined A–A or B–B: $\gamma_m \gamma_n = +1$, so it would *not* flip.

Therefore $\Gamma H \Gamma = -H$ holds **if and only if** H has only A–B entries. This is true for *any* values of the nearest-neighbour couplings, including random ones. In code: `chiral_violation(H)`$= \|\Gamma H \Gamma + H\|/\|H\|$ is zero exactly when the symmetry holds.

1.3 Consequence 1: eigenvalues come in $\pm b$ pairs

If $H\psi = b\psi$, then $H(\Gamma\psi) = -\Gamma H\psi = -b(\Gamma\psi)$. So $\Gamma\psi$ (flip the sign on every B guide) is a mode with $-b$. The spectrum is mirror-symmetric about $b = 0$, i.e. about β_0. [Derived; Verified-by-test: `test_chiral_partner_has_opposite_eigenvalue`]

1.4 Consequence 2: a lonely mode must sit exactly at $b = 0$ and live on one sublattice

A mode at $b = 0$ is its own partner. We can then choose it to be an eigenvector of Γ, which means it lives *only on A* or *only on B*. A single zero mode cannot leave $b = 0$ under a chiral perturbation, because it has no partner to pair with: moving to $+b$ would require a twin at $-b$.
In simple words: the symmetry acts like a mirror at $b = 0$. A mode sitting exactly on the mirror can't move off it alone; it would need a partner to move with.

1.5 What keeps / breaks the symmetry (key for Days 3–6)

Perturbation	Chiral?	Physical origin
Random nearest-neighbour couplings κ_n	**kept**	gap errors δg
Random on-site detuning $\delta\beta_n$	broken	width errors δw
Uniform shift $\delta\beta$ on all guides	broken (but harmless for relative phase)	global width / index change
Next-nearest-neighbour coupling (A–A, B–B)	broken	strong evanescent tails
End-guide self-shift (Limitation L1)	broken	only one neighbour at the end
Uniform loss $-i\alpha/2$ on all guides	broken (shifts all b by an imaginary constant)	material/scattering loss — Day 5

[Derived; Verified-by-test: `test_symmetry_breaking_perturbations_are_detected`]

2 The winding number

2.1 Picture

From Day 1: $H(k) = d_x\sigma_x + d_y\sigma_y$ with $d_x = \kappa_1 + \kappa_2\cos k$, $d_y = \kappa_2 \sin k$. As k goes once around the zone $(0 \to 2\pi)$, the point (d_x, d_y) draws a **circle of radius κ_2 centred at** $(\kappa_1, 0)$ (Fig. 1a).

- $\kappa_1 < \kappa_2$: the circle goes *around* the origin $\Rightarrow W = 1$.
- $\kappa_1 > \kappa_2$: the circle does *not* enclose the origin $\Rightarrow W = 0$.

2.2 Formula and convention [Derived]

$$W = \frac{1}{2\pi}\oint_0^{2\pi} \frac{d\varphi}{dk}\,dk, \qquad \varphi(k) = \arg\left(d_x + i\,d_y\right) = \arg\left(\kappa_1 + \kappa_2 e^{+ik}\right).$$

Convention: k increases, angle counted anticlockwise. (Using $\arg h = \arg(d_x - id_y)$ instead gives $W = -1$; the sign is a convention, the magnitude is the physics. State your choice.) **Numerically** (`winding_number`): sample k on a closed grid, take each small phase step wrapped into $(-\pi, \pi]$, add them, divide by 2π. The result is an integer to 10^{-15} and does not depend on the grid size (tested with 301 and 2001 points).

2.3 Why it is "topological"

W counts how many times a closed curve goes around a point. You cannot change that count by smoothly bending the curve — *unless the curve passes through the point.* Passing through the origin means $|\boldsymbol{d}| = 0$, i.e. $b = 0$ in the bulk: the **gap closes**. So W can only change if (i) the gap closes, or (ii) chiral symmetry is broken: a $d_z\sigma_z$ term lets the curve leave the plane and be lifted over the origin, and then "winding around the origin" no longer means anything.
In simple words: W is like counting how many times a rubber band is looped around a pole. You can stretch the band any way you like; the count only changes if you cut it (close the gap) or lift it off the pole (break the symmetry).

2.4 W depends on how you choose the unit cell

If you choose the cell as (B_n, A_{n+1}) instead of (A_n, B_n), the roles of κ_1 and κ_2 swap, and W flips between 0 and 1 (`test_winding_depends_on_unit_cell_choice`). So W alone is **not** a property of the bulk; it is a property of the bulk *together with a choice of cell.* The edge removes the ambiguity: a real chain is cut in a definite place, and that cut picks the cell. Blanco-Redondo *et al.* use the equivalent *Zak phase* $\gamma_{\text{Zak}} = \pi W$ (0 or π) [Literature].

Viva warning. "The trivial phase has $W = 0$" is only true once you say which cell you used. The physically meaningful, cell-independent statement is about the *difference* between two sides of an interface, or about a given termination.

3 Bulk–boundary correspondence

Statement (for chiral 1-D chains). For a chain cut so that its end cell is the cell used to compute W, the number of zero modes at that end equals $|W|$ (protected by chiral symmetry and the gap). [Literature: Ozawa *et al.* 2019 review — read §on 1-D SSH and note the exact wording]

Our numerical check. For 77 values of κ_1/κ_2 in $[0.05, 3]$ (skipping the gap-closing region near 1), we counted in-gap modes ($|b| < |\kappa_1 - \kappa_2|/2$) of a 200-guide open chain and divided by 2 (two ends). Count per edge $= W$ in **77/77** cases (Fig. 1b). [Simulated; Verified-by-test: `test_bulk_boundary_even_chain`]

4 The edge state and its localisation length

4.1 Derivation (semi-infinite chain starting on A) [Derived]

Look for a mode with $b = 0$ that lives only on A ($b_n = 0$). The A-equations of Day 1, $0 = \kappa_1 b_n + \kappa_2 b_{n-1}$, are then automatic. The B-equations give

$$0 = \kappa_1 a_n + \kappa_2 a_{n+1} \quad \Longrightarrow \quad a_{n+1} = -\frac{\kappa_1}{\kappa_2}\, a_n = -r\, a_n \quad \Longrightarrow \quad a_n = a_1(-r)^{n-1}.$$

- It is a real, normalisable (finite-power) mode only if $|r| < 1$, i.e. $\kappa_1 < \kappa_2$: the end guide hangs on a **weak** bond. This is $W = 1$. ✓
- The sign alternates from one A guide to the next (π phase jump between neighbouring A guides).
- Normalisation: $\sum |a_n|^2 = |a_1|^2/(1 - r^2) = 1 \Rightarrow |a_1|^2 = 1 - r^2$.

4.2 Localisation length

Write $|a_n| = |a_1|e^{-(n-1)/\xi}$. Since $|a_n| = |a_1|r^{n-1}$:

$$\boxed{\xi = \frac{1}{\ln(\kappa_2/\kappa_1)} \text{ unit cells (amplitude)}} \qquad \text{power decays as } e^{-2(n-1)/\xi}, \text{ i.e. by } r^2 \text{ per cell.}$$

In waveguides: one cell = two guides, so the amplitude decay length is 2ξ guides. $\xi \to \infty$ as $r \to 1$ (gap closes, mode spreads into the bulk). $\xi \to 0$ as $r \to 0$ (end guide fully isolated). Values (Fig. 1c): $r = 1/3$: $\xi = 0.91$; $r = 0.5$: $\xi = 1.44$; $r = 0.8$: $\xi = 4.48$ cells. Fit of the *numerical* eigenvector of a 201-guide chain agrees to $< 2 \times 10^{-13}$ (relative). [Derived; Simulated; Verified-by-test]
In simple words: the edge light leaks into the array one strong bond at a time, and each step costs a factor κ_1/κ_2 in amplitude. The weaker κ_1 is compared with κ_2, the more tightly the light hugs the edge.

5 Odd chains: an exact zero mode for ANY nearest-neighbour couplings

Take $M = 2N + 1$ guides (A at both ends). Let each coupling be different, $c_1, c_2, \ldots, c_{2N}$ (e.g. disordered). Repeat Section 4.1 with these couplings:

$$a_{j+1} = -\frac{c_{2j-1}}{c_{2j}}\, a_j \quad (j = 1, \ldots, N), \qquad b_j = 0.$$

This satisfies *every* equation of the finite chain exactly, so $b = 0$ **is an exact eigenvalue for any positive couplings** [Derived; Verified-by-test with 20 random chains, $|b| < 10^{-12}$].

Counting argument (why it must exist). The chain has $N + 1$ A guides and N B guides. Write $H = \begin{pmatrix} 0 & C \\ C^T & 0 \end{pmatrix}$ in (A, B) order; C^T maps the $(N + 1)$-dimensional A space into the N-dimensional B space, so it must send at least one A vector to zero. That vector is a zero mode. This needs only "A couples only to B" — i.e. chiral symmetry — not any particular coupling values.

5.1 What disorder does to localisation [Derived formula; ensemble behaviour = Hypothesis H2]

Taking logs: $\ln|a_{n+1}| = \ln|a_1| + \sum_{j \le n} \ln\!\left(c_{2j-1}/c_{2j}\right)$. For many random cells, the sum grows like $n\,\langle \ln(\kappa_1/\kappa_2)\rangle$ plus fluctuations of size $\sim \sqrt{n}$. So the *typical* decay length is

$$\xi_{\text{typ}}^{-1} = \big\langle \ln \kappa_2 \big\rangle - \big\langle \ln \kappa_1 \big\rangle.$$

With multiplicative disorder $\kappa \to \kappa e^{\sigma g}$ (g zero-mean), $\langle \ln \kappa \rangle$ is unchanged, so **H2: the zero mode stays exactly at $b = 0$ and keeps the same typical ξ, but its profile fluctuates from sample to sample, increasingly far from the edge.** To be tested on Day 4.

Design decision D1. The main study uses **odd** chains (one exact zero mode). Even chains (next section) have *two* edge modes that are not exactly at zero; using them would mix "pinning" with a finite-size effect.

6 Even chains: the two edge modes hybridise

An even chain ($2N$ guides) has a left mode L (on A) and a right mode R (on B). Each is a zero mode of a half-infinite chain, but in a finite chain their tails touch and they mix into $(L \pm R)/\sqrt{2}$ with $b = \pm|t|$, where $t = \langle L|H|R\rangle$ is their overlap coupling. **Calculation** [Derived]. L: $a_n = \sqrt{1 - r^2}(-r)^{n-1}$. R is the mirror image on B: $b_{N+1-m} = \sqrt{1 - r^2}(-r)^{m-1}$. Then $t = \sum_n a_n(\kappa_1 b_n + \kappa_2 b_{n-1})$. For $n \ge 2$ the bracket is zero (it is exactly R's own zero-mode condition). Only $n = 1$ survives, where b_0 does not exist: $t = a_1\kappa_1 b_1 = \kappa_1(1 - r^2)(-r)^{N-1}$, so

$$\boxed{|b| \simeq \kappa_2\,(1 - r^2)\,r^N}$$

Check: Day 1 found $|b|/\bar{\kappa} = 9.537\times10^{-7}$ for $r = 1/2$, $N = 20$, $\kappa_2 = 4/3$: formula gives $\frac{4}{3}\cdot\frac{3}{4}\cdot 2^{-20} = 9.537\times10^{-7}$. ✓ Day-1 hypothesis E6 is now derived. Fig. 1d: agreement better than 10^{-4} once $r^N \ll 1$ (it is a first-order result, so it fails for short chains or $r \to 1$). [Verified-by-test]

7 Physical scale: what the literature actually gives us

From Blanco-Redondo *et al.*, PRL **116**, 163901 (2016) [Literature, read from the author-accepted manuscript]:

- Silicon-on-insulator waveguides, 450 nm × 220 nm, air cladding, $\lambda = 1550$ nm; 203 waveguides; 500 μm long; deep-UV lithography (IMEC).
- Narrow/wide gaps: 166/294 nm ("long-long", topological interface) and 182/324 nm ("short-short", trivial compound defect). Equidistant reference: 230 nm.
- Reported coupling *ratios* ≈ 2.96–3.26 and 0.31–0.34. **Absolute κ values are not given in the text.**
- They use a **domain wall in the middle** of the array (an interface between two SSH chains with Zak phases 0 and π), not an array edge. The topological interface mode kept the same output over an 8 nm wavelength range; the trivial "short-short" defect supported two modes that beat, so its output changed with length and wavelength.

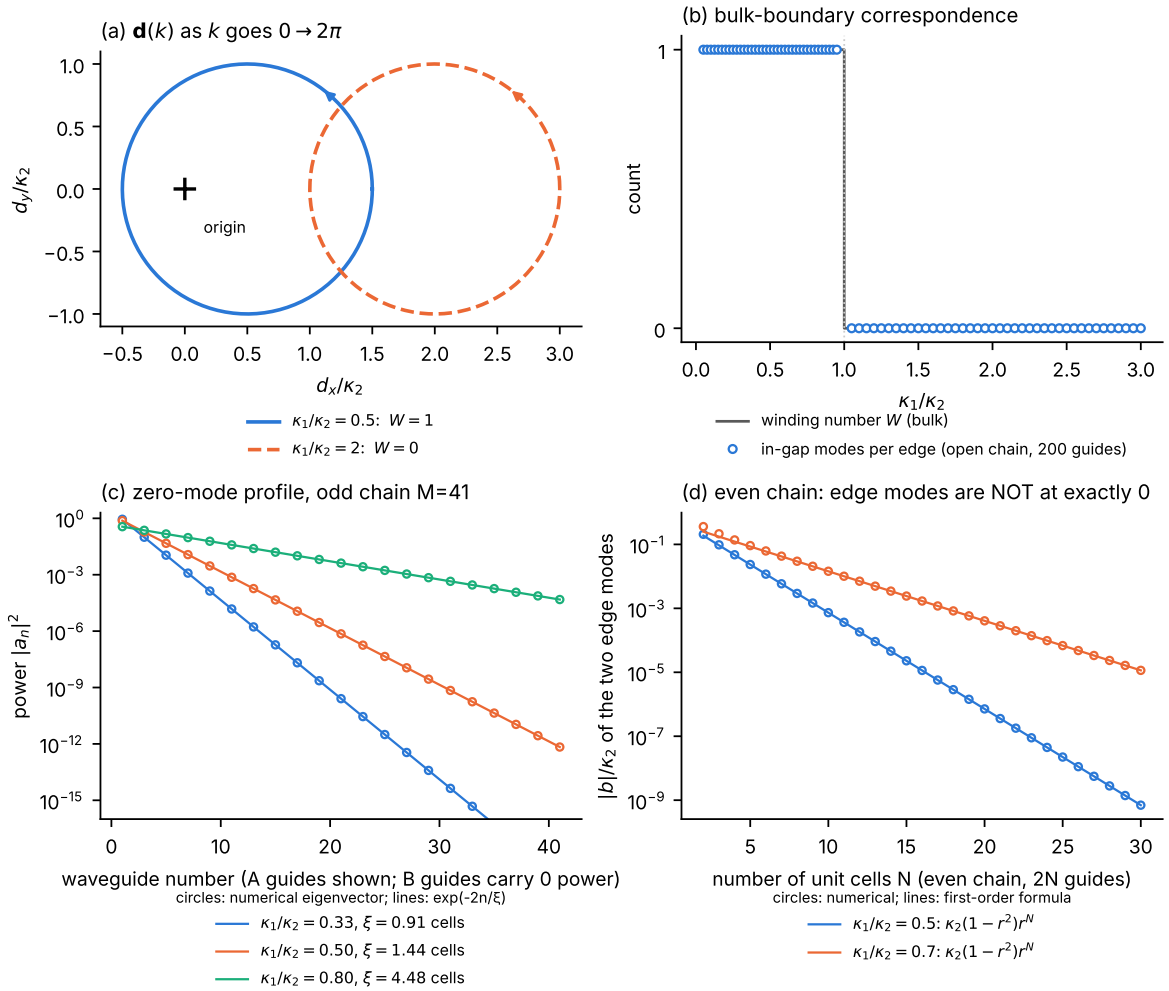


Figure 1: Generated by `scripts/day2_topology.py` from `data/day2_topology.npz` (deterministic). (a) $\boldsymbol{d}(k)$ loops. (b) W vs. in-gap modes per edge of a 200-guide chain. (c) Zero-mode power on the A guides of a 41-guide chain (circles: `eigh`; lines: $e^{-2n/\xi}$); the $r = 1/3$ data stop at 10^{-16} = double-precision round-off. (d) Even-chain edge splitting vs. N.

Decisions. D2: baseline ratio $r = \kappa_1/\kappa_2 = 1/3$ (literature-based). **D3:** keep κ dimensionless (lengths in units of $1/\kappa_2$) until an absolute value is obtained from a cited source or our own mode-solver calculation (optional COMSOL extension). Do not invent a number. **D4 (new, important for Day 3):** their *interface* geometry plus their "short-short" trivial defect is a published, physically fair topological-vs-trivial pair. I recommend we adopt this geometry for the comparison: it launches light into the middle (no end-guide self-shift problem L1), and it has an experimental precedent.

8 Back to the research question: what we have, and what we do not

Concept	Status after Day 2
Eigenvalue pinning	**Derived**: $b = 0$ exactly for chiral perturbations (odd chain), for any strength — even beyond gap closing.
Localisation	**Derived** for the clean chain; disordered behaviour is Hypothesis H2. Needs the gap.
Phase stability	**Not addressed.** $b = 0$ is measured from β_0; the phase is $(\beta_0 + b)L$, and β_0 is not protected.
Transport / output stability	**Not addressed.** A launched field also excites bulk modes (Day 5).
Circuit-level robustness	**Not addressed** (Days 8–10).

9 Code, tests and two lessons

`topq/topology.py`: `chiral_operator`, `chiral_violation`, `winding_number`, `zero_mode_from_couplings`, `edge_mode`, `sublattice_weights`, `ipr`, `fit_localization_length`, `xi_analytic`, `even_chain_splitting`. `tests/test_topology.py`: 31 new tests; whole suite **54 passed**.

Lesson 1 — an "equivalent mutant". In the mutation test I changed Γ to $-\Gamma$ and *no* test failed. That is correct, not a weak test: $(-\Gamma)H(-\Gamma) = \Gamma H \Gamma$, so the overall sign of the chiral operator has no physical meaning. Three other injected bugs (wrong sign in the zero-mode recursion, ξ off by 2, winding direction) were caught.

Lesson 2 — stale cache. After restoring the file, tests briefly still failed: Python reused a cached compiled copy (`__pycache__`) because the restored file had the same size and timestamp second. Deleting `__pycache__` fixed it. If results ever look "impossible" after an edit, clear the cache first.

10 Self-test (answer without notes)

1. Why does a uniform shift $\delta\beta\mathbb{1}$ break chiral symmetry but a random change of every κ_n does not?
2. Draw the $\boldsymbol{d}$-loop for $\kappa_1 = \kappa_2$. Why is W undefined there?
3. An odd chain with $\kappa_1 > \kappa_2$: where is its zero mode, and on which sublattice?
4. Why does the counting argument need "A couples only to B"? Give a perturbation that defeats it.
5. Even chain, $r = 0.5$: how many unit cells do you need for the edge-mode splitting to be below $10^{-6}\kappa_2$?
6. In one sentence: why does $b = 0$ *not* by itself mean the light's phase is protected?